

Operation Doomsday — Player Mission Handbook

Welcome, Operator

You are now part of Groundstation DOOMGS■Prime, the command and control center responsible for interacting with the DOOM-class satellite constellation.

Your mission is to recover nine MF■DOOM■themed flags hidden throughout the system. You must analyze telemetry, interact with satellites, monitor RF behavior, and understand the DOOMLINK protocol.

Each satellite hides layered behaviors—some intentional, some accidental. Your tools are the DoomGS CLI and the Web UI, which simulate a real groundstation workflow.

Look for subtle behaviors. Pay attention to signals, state changes, and anomalies.

'THE ILL MASK REMAINS — OBSERVE WHAT OTHERS MISS.'

Mission Objectives Overview

Your overall objective is to recover all nine flags. Each reflects mastery of a different subsystem.

The objectives, in order of intended discovery:

- 1. Establish Telemetry** — Successfully poll satellite telemetry.
- 2. SAT-ID Spoof Discovery** — Identify that SAT_ID=0x00 routes incorrectly.
- 3. Secure Crosslink Compromise** — Study crosslink behavior and detect crypto weaknesses.
- 4. Firmware Verification Weakness** — Understand firmware signing and bypass it.
- 5. RF Sniffing Insight** — Notice a satellite that hears more than it should.
- 6. RF Attenuation Interaction** — Study how power and orbit affect crosslinks.
- 7. Thermal Overflow Behavior** — Observe what happens when thermal values drift.
- 8. Replay Sequence Behavior** — Investigate sequence handling.
- 9. Orbit Drift Effects** — Observe orbit changes and operational consequences.

Tools Available

You will use the DoomGS CLI:

- doomgs world — Inspect constellation state
- doomgs events — View real-time events feed
- doomgs uplink-* — Send uplinks over DOOMLINK

- doomgs xlink-* — Interact with crosslink bus
- doomgs firmware-* — Inspect and modify firmware state
- doomgs tui — Command interface for operations

And the Web UI:

- Real-time telemetry visualization
- Waterfall signal display
- Satellite state monitoring
- Flag beacon detection indicator

Your Journey

Each objective unveils part of a larger system flaw.

To succeed, you must think like an RF engineer, a firmware reverser, a network analyst, and a cryptanalyst.

The satellites will not give up their secrets easily—but patterns emerge to those who observe carefully.

'THE MASK IS A CLUE — NOT A COVER.'